

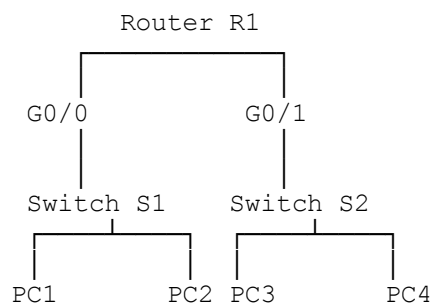
COMPUTER NETWORKS

Ex.No.10	Design a simple topology and configure with one router, two switches and PCs using Cisco Packet Tracer
DATE:	

AIM: To design and configure a simple network topology with one router, two switches, and four PCs in Cisco Packet Tracer for successful inter-network communication.

PROCEDURE:

1. Topology



2. Devices to place

In Cisco Packet Tracer, place:

- 1 Router – Cisco 1941
- 2 Switches – Cisco 2960
- 4 PCs

Use Copper Straight-Through cables.

3. IP Addressing

Device	Interface	IP Address	Subnet Mask	Gateway
R1	G0/0	192.168.1.1	255.255.255.0	—
R1	G0/1	192.168.2.1	255.255.255.0	—
PC1	NIC	192.168.1.10	255.255.255.0	192.168.1.1
PC2	NIC	192.168.1.11	255.255.255.0	192.168.1.1
PC3	NIC	192.168.2.10	255.255.255.0	192.168.2.1
PC4	NIC	192.168.2.11	255.255.255.0	192.168.2.1

So we have **two different networks**:

Network 1: 192.168.1.0/24



Network 2: 192.168.2.0/24



4. Configure Router R1

Click **R1** → **CLI** and enter:

```
enable
configure terminal

interface gigabitEthernet 0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit

interface gigabitEthernet 0/1
ip address 192.168.2.1 255.255.255.0
no shutdown
```

```
exit
```

```
end
```

```
write memory
```

The important command is:

```
no shutdown
```

It activates the router interface.

Check the interfaces using:

```
show ip interface brief
```

You should see the interfaces as **up/up**.

5. Configure PC1

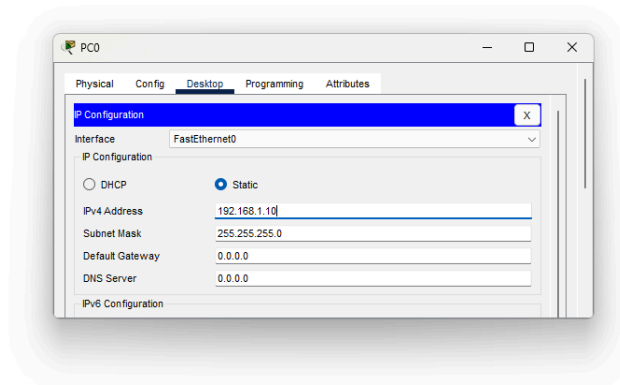
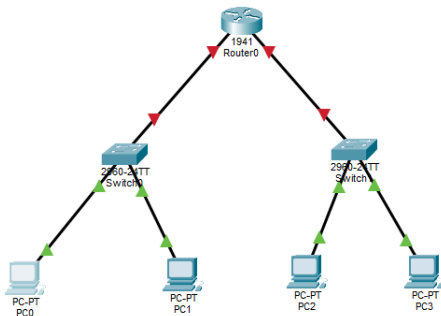
Go to:

PC1 → Desktop → IP Configuration

Enter:

```
IP Address:      192.168.1.10
Subnet Mask:     255.255.255.0
Default Gateway: 192.168.1.1
```

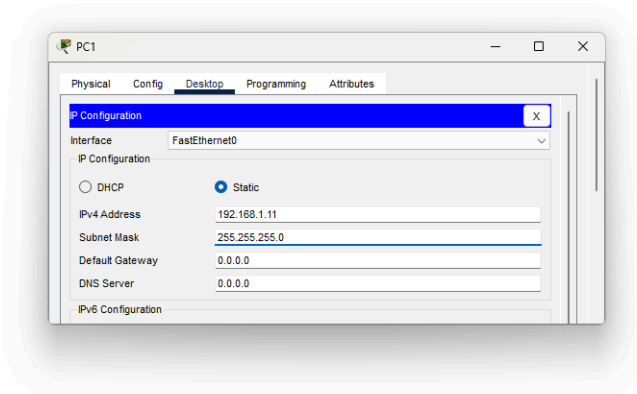
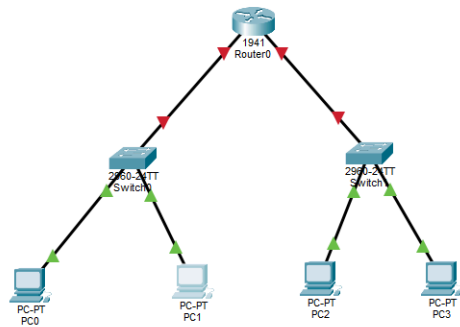
SCREENSHOTS:(to be pasted)



6. Configure PC2

IP Address: 192.168.1.11
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.1.1

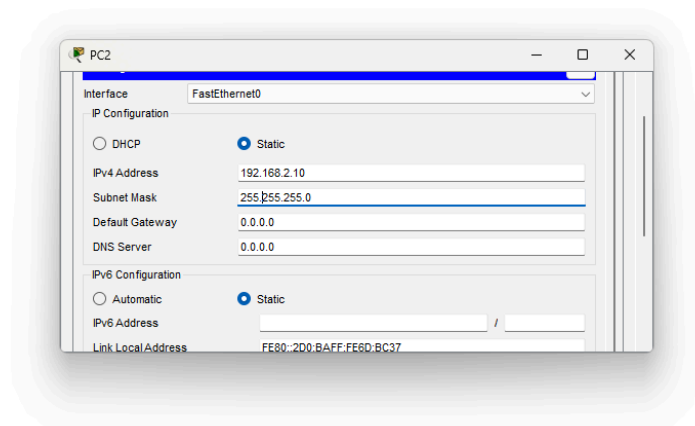
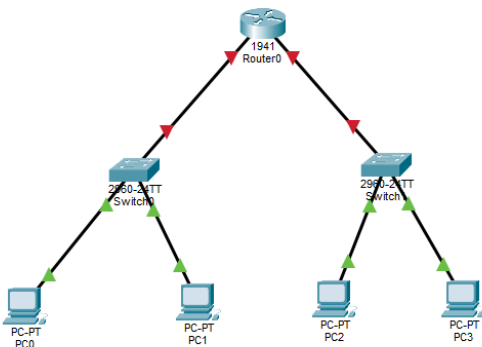
SCREENSHOTS:(to be pasted)



7. Configure PC3

IP Address: 192.168.2.10
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.2.1

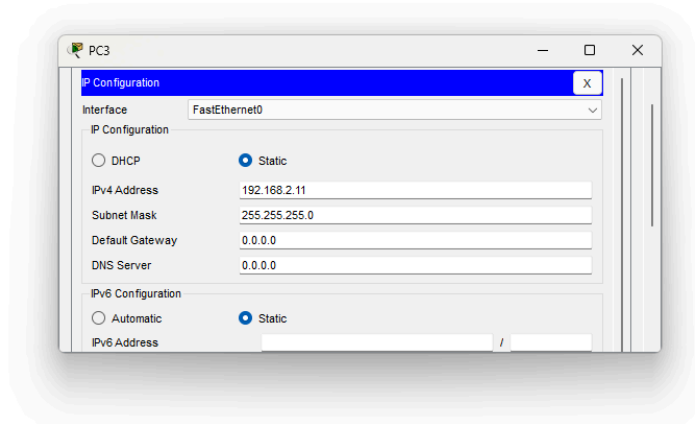
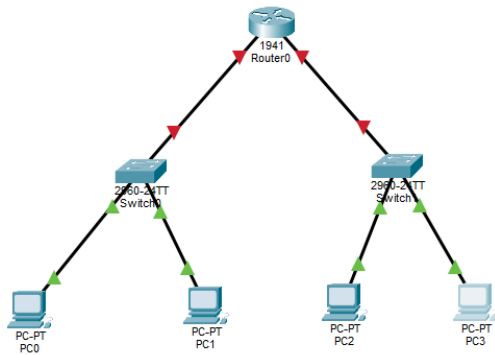
SCREENSHOTS:(to be pasted)



8. Configure PC4

IP Address: 192.168.2.11
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.2.1

SCREENSHOTS:(to be pasted)



9. Test the network

From **PC1** → **Desktop** → **Command Prompt**, type:

```
ping 192.168.1.11
```

This tests **PC1** → **PC2**.

Then:

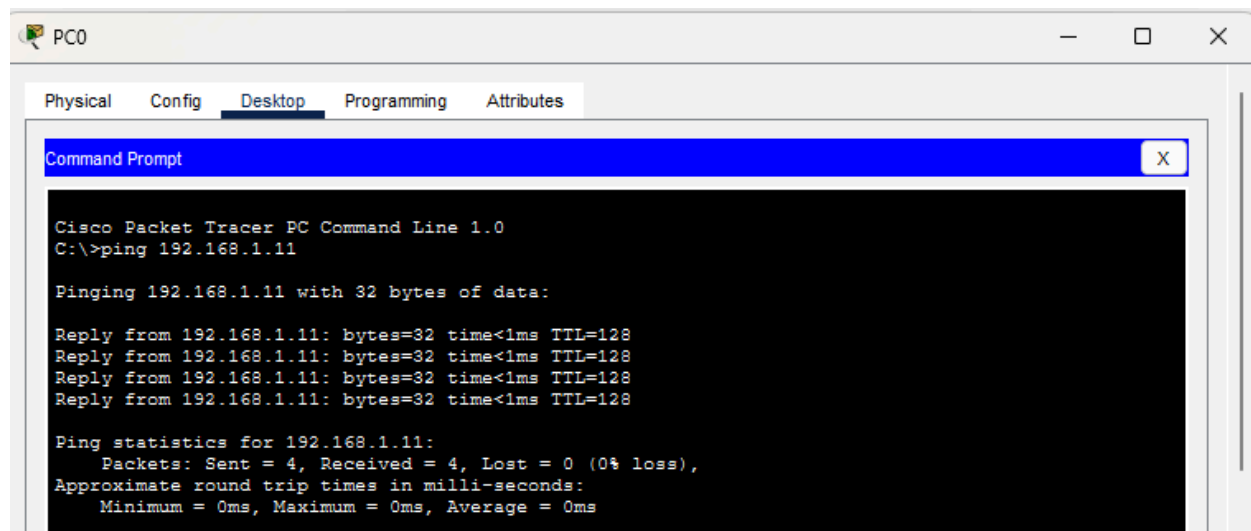
```
ping 192.168.2.10
```

This tests **PC1** → **Router** → **Switch 2** → **PC3**.

You should receive:

```
Reply from 192.168.2.10  
Reply from 192.168.2.10  
Reply from 192.168.2.10  
Reply from 192.168.2.10
```

SCREENSHOTS:(to be pasted)



SAMPLE CONFIGURATION:

TOPOLOGY: One Router, Two Switches and PCs

```
graph TD
    R1((R1)) --- G0/0 --- S1[Switch S1]
    R1 --- G0/1 --- S2[Switch S2]
    S1 --- PC1[PC1]
    S1 --- PC2[PC2]
    S2 --- PC3[PC3]
    S2 --- PC4[PC4]
```

IP ADDRESSING TABLE				
Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	G0/0	192.168.1.1	255.255.255.0	-
R1	G0/1	192.168.2.1	255.255.255.0	-
PC1	NIC	192.168.1.10	255.255.255.0	192.168.1.1
PC2	NIC	192.168.1.11	255.255.255.0	192.168.1.1
PC3	NIC	192.168.2.10	255.255.255.0	192.168.2.1
PC4	NIC	192.168.2.11	255.168.255.0	192.168.2.1

CONFIGURATION STEPS
1. Place Devices

- 1 Router (e.g., 1941)
- 2 Switches (e.g., 2960)
- 4 PCs
- Connections: Copper Straight-Through

2. Configure Router (R1)

```
R1> enable
R1# configure terminal
R1(config)# interface g0/0
R1(config-if)# ip address 192.168.1.1 255.255.255.0
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)# interface g0/1
R1(config-if)# ip address 192.168.2.1 255.255.255.0
R1(config-if)# no shutdown
R1(config-if)# end
R1# write memory
```

3. Configure PCs
For PC1

- IP Address: 192.168.1.10
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.1

For PC2

- IP Address: 192.168.1.11
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.1

For PC3

- IP Address: 192.168.2.10
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.2.1

For PC4

- IP Address: 192.168.2.11
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.2.1

PACKET TRACER TOPOLOGY VIEW

VERIFY CONNECTIVITY
From any PC, open Command Prompt and ping other PCs in the same network and across networks.
Example: From PC1 ping PC3 → ping 192.168.2.10
All pings should be successful.

RESULT: Therefore the process to design and configure a simple network topology with one router, two switches, and four PCs in Cisco Packet Tracer for successful inter-network communication has been simulated, studied and verified.